



Inspiring Excellence

Summer-2025
BTE450: Biotech Project

TOPIC

Isolation and Characterization of Heavy Metal (Lead)-Resistant Bacteria for Bioremediation of Industrial Wastewater

Department of Mathematics and Natural Sciences
BRAC University
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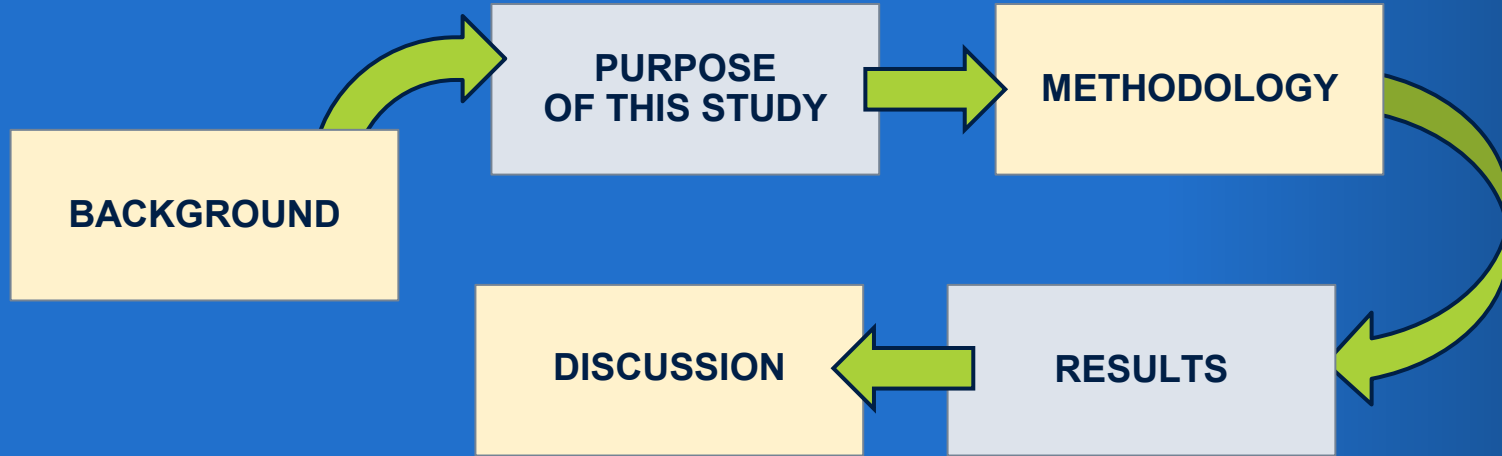
Presented by:

Nazif Muhammad (21236001)

Hasan Mostafiz Tabib (21236043)

Prome Saha (19336004)

SEGMENTS TO BE COVERED IN THIS SESSION



BACKGROUND

Industrial Wastewater & Lead Pollution in Bangladesh

- ❑ Industrial Wastewater is the top leading contributors to environmental degradation.
- ❑ Generated as a by product of various process operations.
- ❑ Contains heavy metals and toxic constituents that release hazardous effects on humans, animals, plants, aquatic, microbial communities and other living organisms on earth.
- ❑ Among all hazardous heavy metals, lead is mostly used highly toxic and persistent compounds.
- ❑ Lead causes greater hazards to health.
- ❑ Lead may result in neurological, cardiovascular and reproductive disorders.

BACKGROUND

Lead Contaminated Sites in Bangladesh by Districts

175

lead-contaminated
sites identified in
Bangladesh

85

battery
recycling
sites

84

smelting industries,
tanneries & heavy
industries

Dhaka and Khulna are the worst-affected districts due to high density of smelting & ULAB industries.

Industrial soils in Bangladesh are severely polluted with lead (**7.3–445 mg/kg**)

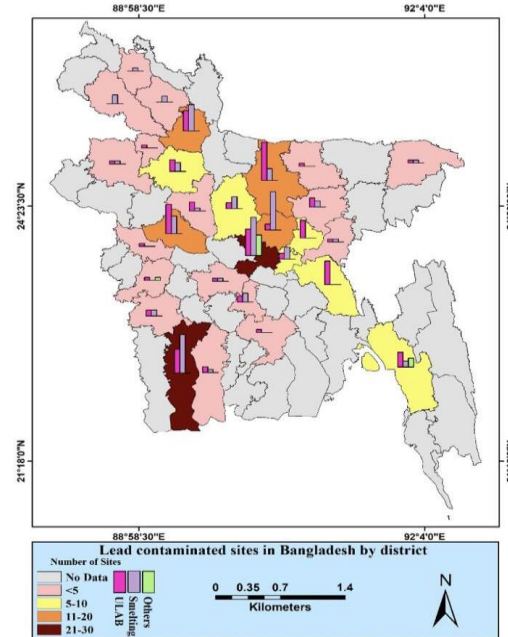


Figure-1: Technological trends in heavy metals removal from industrial wastewater: A review (Majumder et al., 2021)



Figure-2: A lead Battery recycling factory (back) discharging untreated chemical waste (The Battery Boom Driving a Lead Poisoning Epidemic, 2025)

Lead Concentration in Water Sources of Bangladesh

0.04 – 1167 $\mu\text{g/L}$

**Lead in
groundwater**

(safe level: $<50 \mu\text{g/L}$)

0 – 230 $\mu\text{g/L}$

**Lead in surface
water**

(safe level: $<10 \mu\text{g/L}$)

Deep tube wells (average values)

- **0.04 – 1167 $\mu\text{g/L}$ (Groundwater)**
- **0 – 230 $\mu\text{g/L}$ (Surface water)**

Question: Why is lead chosen in the industries?

- Widely used in battery factories and automobile sectors for its unique chemical and physical properties
- Chemically stable as resistant to corrosion and degradation
- Economically realisable as provides long term industrial benefits.
- Less expensive than multiple substitutes
- Suitable for multiple purposes.



Figure-3: Working Environment inside Panna Battery Limited (Panna Group, n.d.)

Limitations of Conventional Approaches



Figure-4: Technological trends in heavy metals removal from industrial wastewater: A review (Shrestha, R.2021)

- **Chemical Precipitation:** Generates large volumes of toxic sludge
- **Coagulation–Flocculation:** Fails to meet regulatory discharge limits
- **Ion Exchange:** Highly expensive due to resin and regeneration chemicals
- **Membrane Separation:** Low persistence

Microbial Compounds: An Effective Alternative to Chemicals for Lead (Pb) Removal

- ❖ Environment friendly and sustainable.
- ❖ Non existence of secondary pollution
- ❖ Specificity and adaptability of bacteria.
- ❖ Restoration of contaminated sites.
- ❖ Economically beneficial.

Microbial Bioremediation of Lead (Pb)

Techniques:

- **Biosorption:** Cell walls/extracellular structures bind Pb using functional groups (carboxyl, amino, phosphate).
- **Efflux:** Bacteria actively pump Pb ions out after reducing toxicity.

Types:

- **In- Situ:** Contaminants treated directly at polluted sites (soil/groundwater).
- **Ex- Situ:** Contaminants removed and treated in controlled environments (example, bioreactors)

Atomic Absorption Spectroscopy (AAS)

- ❖ Determines the wavelengths at which free atoms absorb visible and ultraviolet light.
- ❖ Absorbance compared with calibration curve for concentration.
- ❖ Spectrometer, flame/graphite atomiser, and hollow cathode lamp for radiation.
- ❖ **Lead detection, bacterial bioremediation, soil pollution analysis, and metal quantification in solutions.**
- ❖ Fast and precise analysis, trace-level detection, and high specificity.



Figure-5: Outer Interface of Atomic Absorption Spectroscopy (AAS)

Scanning Electron Microscopy (SEM)

- ❖ **Produces high-resolution images** of sample surfaces using a focused electron beam.
- ❖ Reveals topography, morphology, and composition at nanometer scale.
- ❖ **Provides magnification** up to several hundred thousand times.
- ❖ In bacterial Pb sequestration studies, it can show cell size, shape, and surface texture changes.

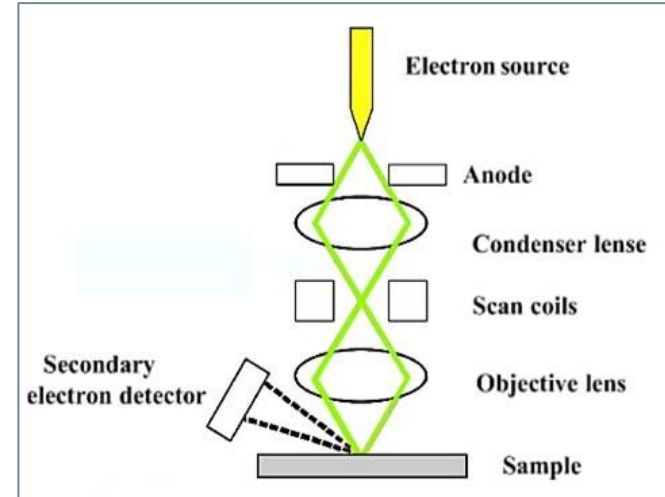


Figure-6: Anatomy of Scanning Electron Microscope (Nanakoudis, 2019)

Purpose of This Study:

- ▶ To isolate lead-resistant bacteria from contaminated industrial soil
- ▶ To assess their metal resistance, and removal efficiency
- ▶ To identify the most effective strains for potential application in wastewater bioremediation

METHODOLOGY

1. Sample Collection

Sample Collection Sites:

Automobile workshop in Mohakhali, Dhaka

Former battery factory area in Comilla

Rationale:

Both sites had long-term exposure to lead (Pb), increasing the likelihood of finding Pb-resistant bacteria.



2. Soil Analysis by AAS

Soil Digestion for AAS:

- Mixed with HNO_3 and HCl
- Heated to dryness, cooled, centrifuged, and diluted

AAS Analysis:

Conducted by BCSIR to quantify Pb levels.

3. Metagenomic Analysis



4. Bacterial Isolation

Metagenomic Sequencing:

- Soil DNA extracted and sequenced by DNA Solutions Ltd

Analysis:

- FASTQ files processed using Porechop and FASTP
- Taxonomic profiling via Kraken and Krona

- Soil dilutions (10^{-1} to 10^{-4}) spread on **Pb(NO₃)₂** supplemented Nutrient Agar
- Incubated at 37°C for 24 hours
- Subcultured on Pb-supplemented agar and broth
- Two isolates labeled: L1 (Comilla) and L2 (Mohakhali)

METHODOLOGY

5. Bacterial analysis with AAS

AAS of Bacterial Supernatants:

- L1 and L2 grown in 100 ppm Pb-supplemented nutrient broth for 24 hours
- Centrifuged to collect cell-free supernatant
- Sent to BCSIR for Pb concentration analysis

6. SEM Sample Preparation

- Bacterial pellets fixed with 2.5% glutaraldehyde
- Gradient dehydration with ethanol (30% to 100%)
- Samples sent to BCSIR for SEM imaging

7. Bacterial Identification

16S rDNA Sequencing:

- Bacterial isolates (Subcultures of L1 and L2) sent to DNA Solutions Ltd
- DNA extracted and sequenced

Bioinformatics Tool:

- BLASTn: Used for species identification

RESULTS

1. AAS of Soil Samples

Confirmed high lead (Pb) contamination in both soil samples.

42.6 ppm Pb
Comilla (battery factory)

61.5 ppm Pb
Mohakhali (Automobile workshop):

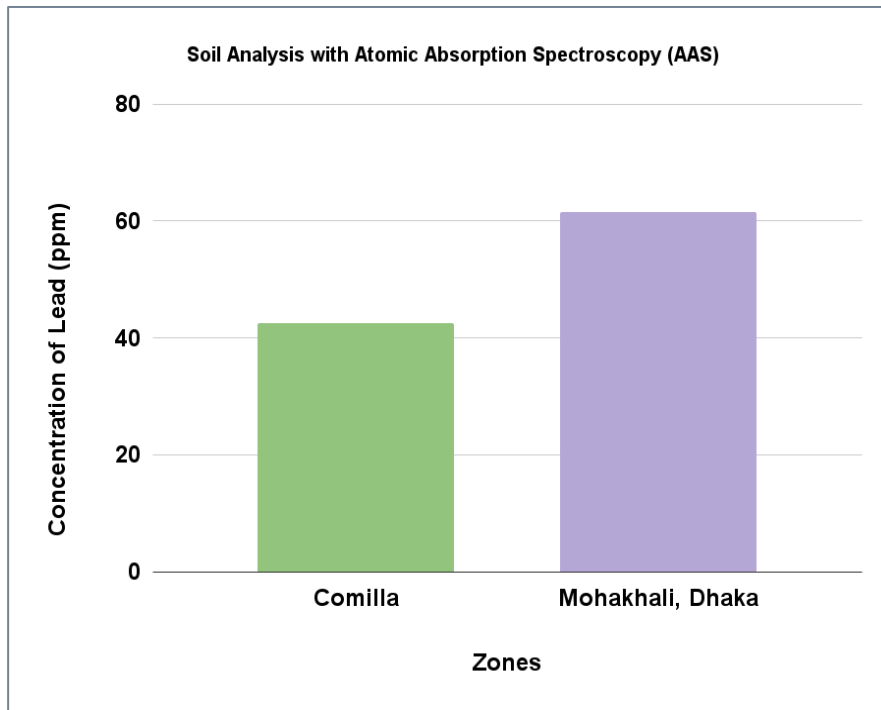


Figure-7: Bar Chart of Soil Analysis of Comilla and Mohakhali, Dhaka soil samples with Atomic Absorption Spectroscopy

2. Metagenomic Analysis of Soil Samples

(i)



RESULTS

2. Metagenomic Analysis of Soil Samples

Revealed a diverse bacterial community adapted to heavy metal stress.

Major Kingdoms:

**80% of Bacteria
Pseudomonadati**

**17% of Bacteria
Bacillati**

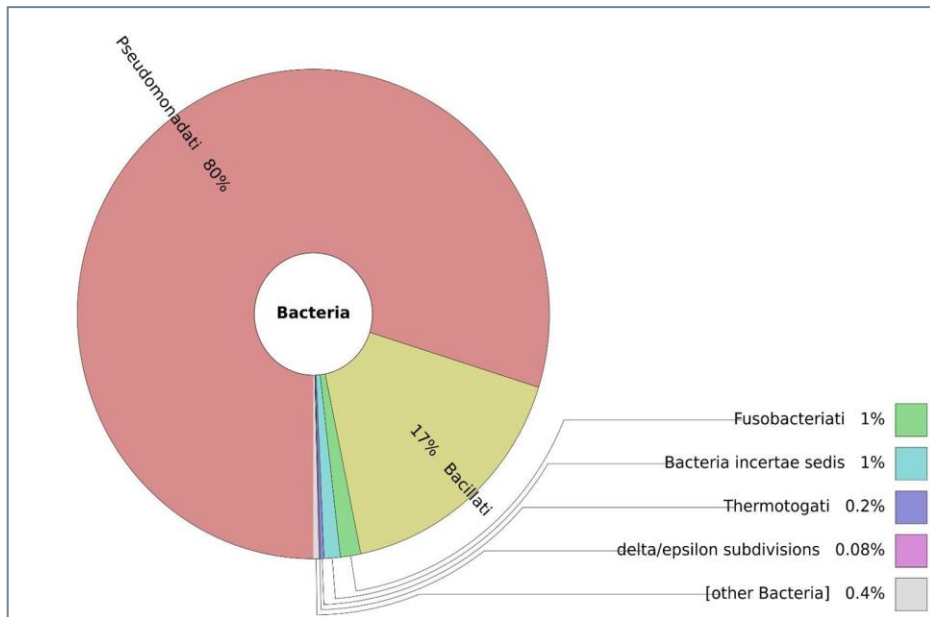


Figure-9: Krona Pie Chart showing bacterial population in dataset 3 (from soil sample of Mohakhali, Dhaka) after metagenomic analysis

RESULTS

2. Metagenomic Analysis of Soil Samples

Revealed a diverse bacterial community adapted to heavy metal stress.

Major Kingdoms:

**73% of Bacteria
Pseudomonadati**

**26% of Bacteria
Bacillati**

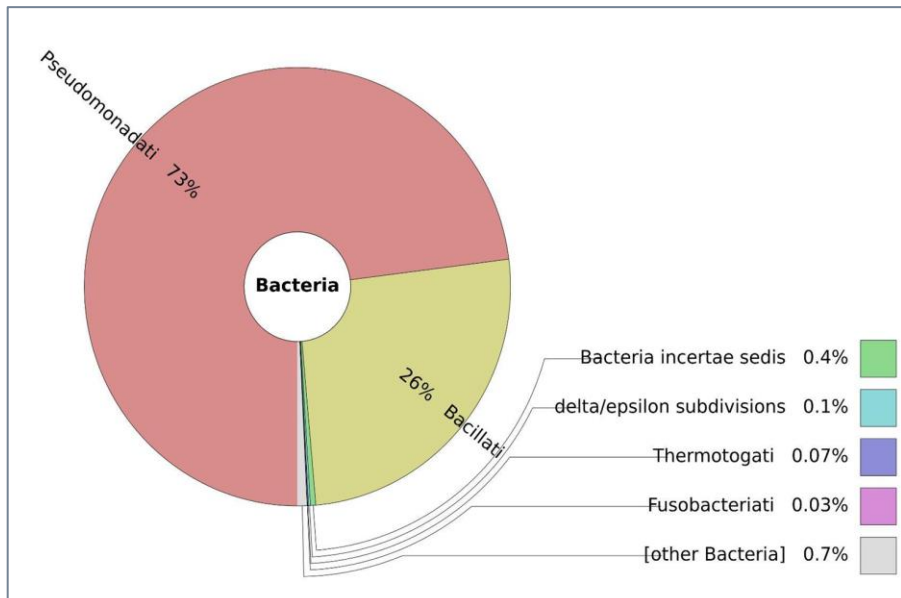


Figure-10: Krona Pie Chart showing bacterial population in dataset 4 (from soil sample of Comilla) after metagenomic analysis

3. Isolation of Bacteria

Two distinct bacterial strains were isolated from the Pb-supplemented agar plates labeled as:

- L1 (from Comilla)
- L2 (from Mohakhali)

4. AAS of Bacterial Samples

Both strains showed remarkable ability to remove lead from solution, confirming their high potential for lead bioremediation.

**L1 removed ~97% of Pb
from a 100 ppm solution**

**L2 removed ~95% of Pb
from a 100 ppm solution**

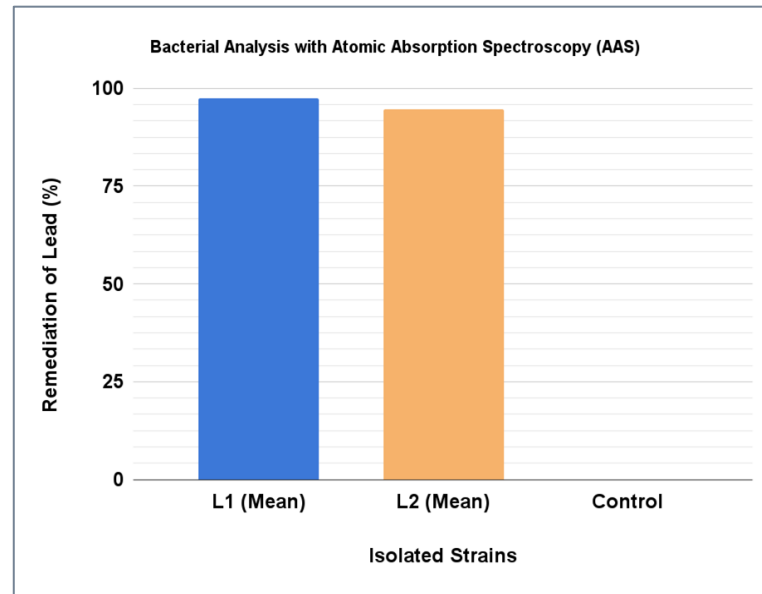


Figure-11: Bar Chart of Bacterial Analysis of Comilla (L1 strain) and Mohakhali, Dhaka (L2 strain) samples with Atomic Absorption Spectroscopy

5. SEM Findings

Control cells (no lead):

Exhibited healthy, rod-shaped cells with smooth, intact surfaces.

Lead-exposed cells: Showed clear signs of stress-induced morphological damage by:

- Surface roughness and deformation
- Cellular aggregation and clustering
- Irregular shapes

These physical changes are likely due to Pb binding and uptake, demonstrating a direct interaction between the bacteria and the heavy metal.

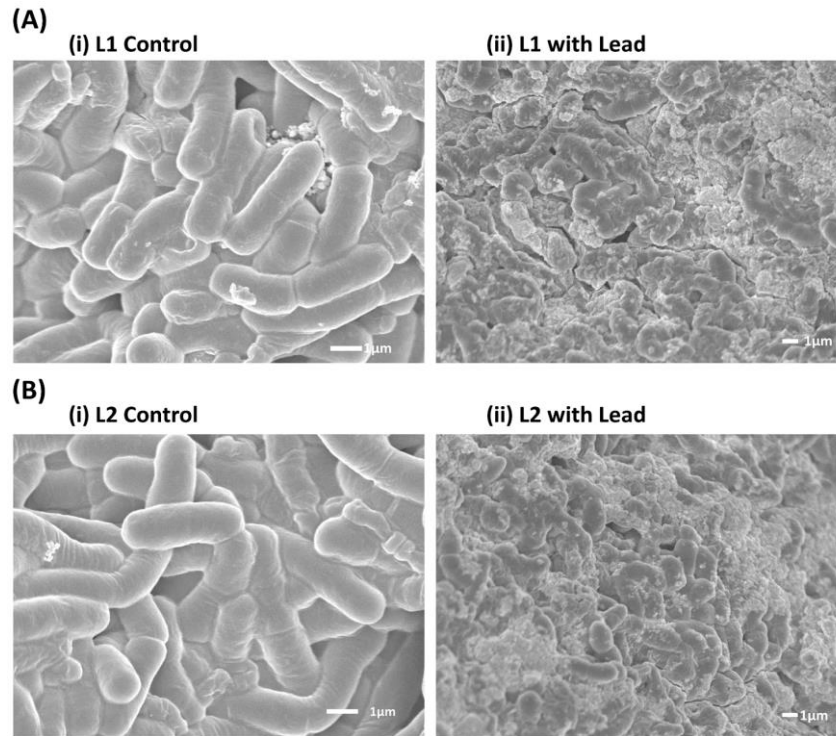


Figure-12: SEM images of bacterial morphology before and after lead exposure.

6. Identification of Microbes

- The data of **16S rDNA sequencing** of two isolated strains (L1 and L2) were obtained from DNA Solutions Ltd as a **FASTA file** format.
- **BLASTn** results showed:
 - **L1** having **98.45%** of percent identity match, and
 - **L2** having **98.97%** of percent identity match with the genus *Priestia*.
- The ***Priestia spp.*** bacteria **belong to Bacillati** kingdom (as explored in prior metagenomic analysis of soil samples.)

DISCUSSION

Soil Sample Analysis:

- ❑ Bangladesh has poorly planned urban areas – works as a persistent sink for Pb (Santos et al., 2025)
- ❑ Comilla (former battery factory, 42.6 ppm Pb) and Mohakhali (automobile workshop with painting and battery dumping, 61.5 ppm Pb) sites were selected.
- ❑ Criteria for sites – chronic exposure to Pb.
- ❑ AAS was selected for its high sensitivity and specificity – outperforming biochemical tests.
- ❑ The environmental stress of soil samples were developed in the lab setting by supplementing nutrient agar and nutrient broth later with 100 ppm of lead from the source, $\text{Pb}(\text{NO}_3)_2$.

Soil Sample Analysis:

- ❑ Nutrient Agar was used to grow bacterial colonies and for subculturing of the bacterial colonies.
- ❑ NA supplemented with Pb ensures the growth of bacterial colonies capable of tolerance or detoxification (Sridevi et al., 2018).
- ❑ Using NB supplemented with lead nitrate will ensure uniform exposure of Pb at various concentrations to the bacteria.
- ❑ 100 ppm was chosen as the threshold based on the assessments of the soil samples.

Bacterial Analysis:

- ❑ The standard database for bacteria was used for bacterial analysis.
- ❑ The isolate L1 and L2 were specified as *Priestia* spp. via 16S rDNA sequencing 98.45% and 98.97% of percent identity in BLASTn, respectively
- ❑ *Priestia* spp. demonstrated high Pb removal capability (approximately 78 ppm was removed, 2 ppm was available in 80 ppm of available lead in liquid broth)
- ❑ SEM test results:
 - Control isolate showed smooth, intact rod-shaped cells.
 - Pb-resistant *Priestia* spp. cells rough surfaces with deformity, and disruptions, proof for Pb binding and uptake have caused stress-induced structural damage.

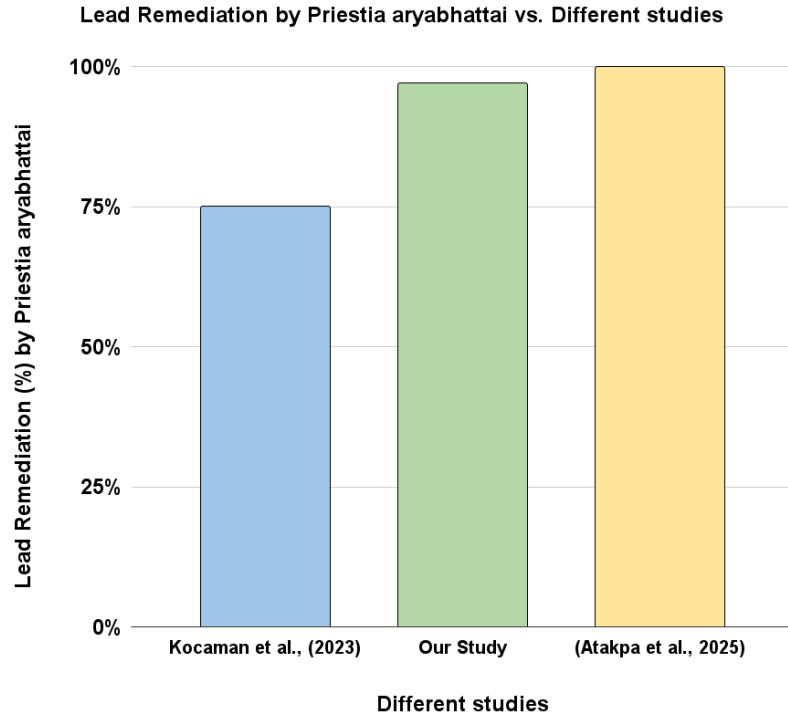


Figure-13: Overall comparison of remediation of lead (Pb) by bacteria, where Study 1 had 75% remediation rate whereas study had 98.98% (near perfect remediation of Pb)

Overall Comparison: Remediation of lead (Pb) by bacteria

Study 1:

Kocaman, A., Savaş, B. F., & İşler Ceyhan, D. (2023). Removal efficacy of toxic metals in leachate through micro-organisms isolated from the natural environment. *BioResources*, 18(3), 5476–5493.

<https://doi.org/10.15376/biores.18.3.5476-5493>

Study 2:

Atakpa, E. O., Li, J., Dong, F., Lei, W., Lei, C., Yunyi, Z., Ke, H., Suanon, F., Sesu, F., Zhou, L., Zhenzhen, L., Tingting, H., & Han, Y. (2025). Deciphering the biomolecular mechanisms of calcite precipitation of multi-metals by the novel metallotolerant strain *Priestia aryabhatai* PMASS1. *Journal of Environmental Chemical Engineering*, 13(5), 117703.

<https://doi.org/10.1016/j.jece.2025.117703>

Metagenomic analysis on Comilla Soil Sample:

- ***Priestia* spp.** is part of the **Bacillati** kingdom.
- Abundance of Firmicutes, Proteobacteria, and Actinobacteria abundance, indicating the presence of acidic, metal-rich conditions from battery factory waste.
- In the enriched environment before the stress conditions, the presence of *Priestia* spp. was moderate (overall 5%)
- *Priestia megaterium* can precipitate Pb into an insoluble form (Pouchucq et al., 2025). Using this method, they help promote the growth factor of plants.
- But the research proved that *Priestia aryabhatai* is capable of remediating Pb.
- The survival of *Priestia* spp. indicates the factory area in Comilla is heavily polluted, suggesting constant Pb exposure, such as water leaching. It also shows the deadly effects of Pb pollution because, before that, a rich microbial community was present.

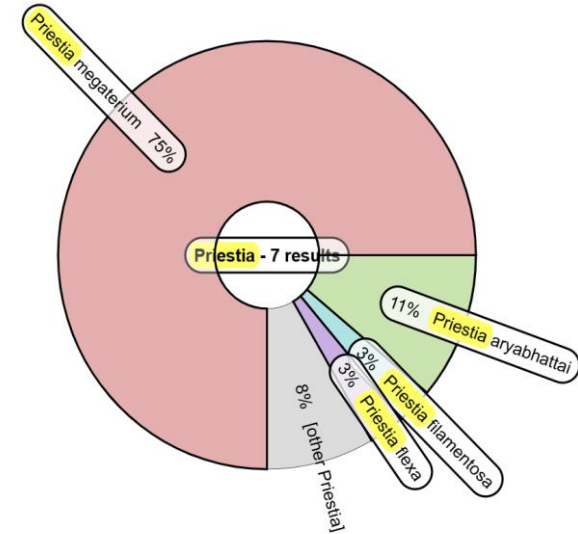


Figure-14: Krona Pie Chart showing overall presence of *Priestia* spp., *Priestia megaterium* is dominant with an 75% reading, but the *Priestia aryabhatai* reading is 11%.

CONCLUSION

Achievement

Isolated *Priestia* spp. strains removing 95-97% of lead from contaminated liquid solutions, demonstrating exceptional efficiency for treating industrial wastewater.

Unlocking Sustainable Alternative

Microbial bioremediation can be a sustainable process for practical application in bioremediation.

Pathway to Implementation

Proposing about the development of controlled bioreactor systems to harness this technology for faster, more manageable, and scalable wastewater treatment.

REFERENCES

- Abbas, S., Ahmed, I., Iida, T., Lee, Y.-J., Busse, H.-J., Fujiwara, T., & Ohkuma, M. (2015). A heavy-metal tolerant novel bacterium, *Alcaligenes pakistanensis* sp. nov., isolated from industrial effluent in Pakistan. *Antonie van Leeuwenhoek*, 108(4), 859–870. <https://doi.org/10.1007/s10482-015-0540-1>
- Abba, M., Shafiu, S., Ibrahim, B., Aisha, M., Aliyu, H., Jamilu, Y., & Umar, Y. M. (2024). Understanding the Impact of Lead Contamination on Microbial Diversity: A Comprehensive Review. *International Journal of Research Publication and Reviews*, 5(7), 1119–1130. <https://doi.org/10.55248/gengpi.5.0724.1721>
- Atakpa, E. O., Li, J., Dong, F., Lei, W., Lei, C., Yunyi, Z., Ke, H., Suanon, F., Sesu, F., Zhou, L., Zhenzhen, L., Tingting, H., & Han, Y. (2025). Deciphering the biomolecular mechanisms of calcite precipitation of multi-metals by the novel metallotolerant strain *Priestia aryabhattai* PMASS1. *Journal of Environmental Chemical Engineering*, 13(5), 117703. <https://doi.org/10.1016/j.jece.2025.117703>
- Beaty, R., & Kerber, J. (1993). Concepts, Instrumentation and Techniques in Atomic Absorption Spectrophotometry Second Edition. *THE PERKIN-ELMER CORPORATION*. https://www.iitk.ac.in/che/PG_research_lab/pdf/resources/AASGTA-reading-material.pdf
- Chowdhury, I. R., Chowdhury, S., Mazumder, M. A. J., & Al-Ahmed, A. (2022). Removal of lead ions (Pb²⁺) from water and wastewater: a review on the low-cost adsorbents. *Applied Water Science*, 12(8). <https://doi.org/10.1007/s13201-022-01703-6>

- Fischer, E. R., Hansen, B. T., Nair, V., Hoyt, F. H., Schwartz, C. L., & Dorward, D. W. (2024). *Scanning Electron Microscopy. Current Protocols*, 4(5). <https://doi.org/10.1002/cpz1.1034>
- George, S. E., & Wan, Y. (2019). Advances in characterizing microbial community change and resistance upon exposure to lead contamination: Implications for ecological risk assessment. *Critical Reviews in Environmental Science and Technology*, 50(21), 2223–2270. <https://doi.org/10.1080/10643389.2019.1698260>
- Harun, F. A., Yusuf, M. R., Usman, S., Shehu, D., Babagana, K., Sufyanu, A. J., Jibril, M. M., Bello, A. M., Musa, K. A., Jagaba, A. H., Shukor, M. Y., & Yakasai, H. M. (2023). Bioremediation of lead contaminated environment by *Bacillus cereus* strain BUK_BCH_BTE2: Isolation and characterization of the bacterium. *Case Studies in Chemical and Environmental Engineering*, 8, 100540. <https://doi.org/10.1016/j.cscee.2023.100540>
- Hložková, K., Šuman, J., Strnad, H., Ruml, T., Paces, V., & Kotrba, P. (2013). Characterization of pbt genes conferring increased Pb²⁺ and Cd²⁺ tolerance upon *Achromobacter xylosoxidans* A8. *Research in Microbiology*, 164(10), 1009–1018. <https://doi.org/10.1016/j.resmic.2013.10.002>
- Hu, C., Yang, Z., Chen, Y., Tang, J., Zeng, L., Peng, C., Chen, L., & Wang, J. (2024). Unlocking soil revival: the role of sulfate-reducing bacteria in mitigating heavy metal contamination. *Environmental Geochemistry and Health*, 46(10). <https://doi.org/10.1007/s10653-024-02190-1>
- Kocaman, A., Savaş, B. F., & İşler Ceyhan, D. (2023). Removal efficacy of toxic metals in leachate through micro-organisms isolated from the natural environment. *BioResources*, 18(3), 5476–5493. <https://doi.org/10.15376/biores.18.3.5476-5493>

- Li, Y., Ali, A., Jeyasundar, P. G. S. A., Azeem, M., Tabassum, A., Guo, D., Li, R., Mian, I. A., & Zhang, Z. (2021). *Bacillus subtilis* and saponin shifted the availability of heavy metals, health indicators of smelter contaminated soil, and the physiological indicators of *Symphytum officinale*. *Chemosphere*, 285, 131454. <https://doi.org/10.1016/j.chemosphere.2021.131454>
- Majumder, A. K., Nayeem, A. A., Islam, M., Akter, M. M., & Carter, W. S. (2021). Critical review of lead pollution in Bangladesh. *Journal of Health and Pollution*, 11(31). <https://doi.org/10.5696/2156-9614-11.31.210902>
- Naik, M. M., & Dubey, S. K. (2013). Lead resistant bacteria: Lead resistance mechanisms, their applications in lead bioremediation and biomonitoring. *Ecotoxicology and Environmental Safety*, 98, 1–7. <https://doi.org/10.1016/j.ecoenv.2013.09.039>
- Naik, M., Dubey, S. K., Khanolkar, D. S., & D'Costa, B. (2013). P-type ATPase and MdrL efflux pump-mediated lead and multi-drug resistance in estuarine bacterial isolates. *Current Science*, 105(10), 1366–1372. https://www.researchgate.net/publication/285715434_Ptype_ATPase_and_MdrL_efflux_pump-mediated_lead_and_multidrug_resistance_in_estuarine_bacterial_isolates
- Naik, M. M., Shamim, K., & Dubey, S. K. (2012). Biological characterization of lead resistant bacteria to explore role of bacterial metallothionein in lead resistance. *Current Science*, 103(23), 1113–1130. https://www.researchgate.net/publication/231705552_Biological_characterization_of_lead_resistant_bacteria_to_explore_role_of_bacterial_metallothionein_in_lead_resistance

- Navgire, G. S., Goel, N., Sawhney, G., Sharma, M., Kaushik, P., Mohanta, Y. K., Mohanta, T. K., & Al-Harrasi, A. (2022). Analysis and Interpretation of metagenomics data: an approach. *Biological Procedures Online*, 24(1). <https://doi.org/10.1186/s12575-022-00179-7>
- Nanakoudis, A. (2019, November 15). What is SEM? Scanning Electron Microscopy Explained. *Advancing Materials*. <https://www.thermofisher.com/blog/materials/what-is-sem-scanning-electron-microscopy-explained/>
- Panna Battery Limited – Panna Group. (n.d.). <https://www.pannagroupbd.com/concerns/panna-battery-limited/>
- Park, J. H., Bolan, N., Megharaj, M., & Naidu, R. (2011). Isolation of phosphate solubilizing bacteria and their potential for lead immobilization in soil. *Journal of Hazardous Materials*, 185(2-3), 829–836. <https://doi.org/10.1016/j.jhazmat.2010.09.095>
- Pouchucq, L., Marcoleta, A. E., Rojas, C., Arce, H., Becerra, C., Bahamondes, C., & Lobos Ruiz, P. (2025). Genomic and Functional Characterization of *Priestia megaterium* MOD5IV isolated from the rhizosphere of *Caesalinea spinosa* (Mol.): Enhancing Phytoremediation Potential in Multi-Metal Contaminated Soils. *BioRxiv (Cold Spring Harbor Laboratory)*. <https://doi.org/10.1101/2025.04.15.648928>
- Raj, K., & Das, A. P. (2023). Lead pollution: Impact on Environment and Human Health and Approach for a Sustainable Solution. *Environmental Chemistry and Ecotoxicology*, 5, 79–85. <https://doi.org/10.1016/j.eneco.2023.02.001>
- Shrestha, R., Ban, S., Devkota, S., Sharma, S., Joshi, R., Tiwari, A. P., Kim, H. Y., & Joshi, M. K. (2021). Technological trends in heavy metals removal from industrial wastewater: A review. *Journal of Environmental Chemical Engineering*, 9(4), 105688. <https://doi.org/10.1016/j.jece.2021.105688>

Sridevi, V., Raghuram, M., & Ravisankar, D. (2018). Isolation and Screening of Heavy Metal Resistant Microorganisms From Industrial Soil. *Journal of Pure and Applied Microbiology*, 12(3), 1667–1674. <https://doi.org/10.22207/jpam.12.3.75>

SURFACE AND GROUND WATER QUALITY REPORT 2021. Department of Environment Ministry of Environment, Forest & Climate Change Government of the People's Republic of Bangladesh Draft Final Report. (n.d.). https://doe.gov.bd/sites/default/files/files/doe.portal.gov.bd/publications/e8dbdadb_7504_44e6_851d_bb4ca6b108c8/2024-08-27-07-59-94819bf17dac3a9fbdcbcf8f48fc6aa68.pdf

The battery boom driving a lead poisoning epidemic. (2025, April 1). *New Vision*.

https://www.newvision.co.ug/category/report/the-battery-boom-driving-a-lead-poisoning-epi-NV_207979

Volaric, A., Svircev, Z., Tamindzija, D., & Radnovic, D. (2021). Microbial bioremediation of heavy metals. *Chemical Industry*, 75(2), 103–115. <https://doi.org/10.2298/hemind200915010v>

Wood, D. E., & Salzberg, S. L. (2014). Kraken: Ultrafast metagenomic sequence classification using exact alignments. *Genome Biology*, 15(3), R46. <https://doi.org/10.1186/gb-2014-15-3-r46>

Zhang, Y., Peng, J., Wang, Z., Zhou, F., Yu, J., Chi, R., & Xiao, C. (2025). Metagenomic analysis revealed the bioremediation mechanism of lead and cadmium contamination by modified biochar synergized with *Bacillus cereus* PSB-2 in phosphate mining wasteland. *Frontiers in Microbiology*, 16. <https://doi.org/10.3389/fmicb.2025.1529784>

Thank you

for staying with us throughout the presentation!